A01-34196

AIAA-2001-3447

Modern SRM Ignition Transient Modeling (part 5): Prospective Developments in CFD Simulation

J. C. T. Wang

The Aerospace Corporation
P. O. Box 92957 M4-965
Los Angeles, CA 90009-2957

37th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit
8 -11 July 2001 / Salt Lake City, Utah

Modern SRM Ignition Transient Modeling (Part 5) Prospective Developments in CFD Simulation

Johnson C. T. Wang*
The Aerospace Corporation
El Segundo, California

Abstract

The current state-of-the-art in solid rocket motor (SRM) ignition transient prediction is reviewed. Although the foundation of the computational fluid dynamics (CFD) approach to SRM ignition transient predictions was laid more than a quarter century ago by Peretz, Kuo, Caveny, and Summerfield, rapid advances have been made only recently. Three-dimensional (3-D) prediction tools are being developed and may require the usage of parallel computing systems to alleviate displeasingly long computer run times. In this paper, a quasi-3D CFD approach is presented as an alternative cost-saving method for 3-D ignition transient flow predictions. Justifications and limitations are discussed. The approach is applied to two generic SRMs to illustrate: (1) 3-D flow phenomena that can not be obtained by using a 1-D or 2-D approach and (2) an easy route that leads to parallel CFD.

Introduction

Depending on the application objective, solid rocket motor (SRM) ignition transient phenomena can be analyzed using methods with different degrees of computational sophistication. Currently, the analytic tools available for practical applications range from the simple volume-filling method to the complicated multi-disciplinary axi-symmetric computational fluid dynamics (CFD) codes. The volume-filling method is a simple procedure for obtaining a transient chamber pressure and can be used for motor casing preliminary design.[1] The method had been considered appropriate only for motors of small length-to-diameter (L/D) ratios. However, it was reported recently by Luke et al[2] that the volume-filling method was applied to compute the chamber pressure of the Space Shuttle's Redesigned Solid Rocket Motor (RSRM), which has a very large L/D ratio. The RSRM was modeled as two inter-connected volumes with the volume-filling method applied accordingly. The predicted chamber pressure rise rate was in good agreement with that obtained by using a 1-D CFD code. If the details of flow physics are needed, such as in the

* Senior Engineering Specialist

case of dealing with the effects of propellant cracks on the SRM performance,[4] then a flowfield solution coupled with heat transfer and ignition is required. The foundation for CFD flowfield approaches to the SRM ignition transient prediction was laid by Peretz, Kuo, Caveny, and Summerfield[2] more than a quarter century ago. However, rapid advances have been made only recently.

In the 1970s and 1980s, the progress of analytical predictions of SRM ignition transients was hampered by the lack of a reliable computational method for solving compressible flow equations and the insufficiency of available computing power. As will be illustrated later in this paper, the SRM ignition transient flowfields are dominated by the compressible flow wave interactions. The singular nature of the strong compressions and shocks in the compressible flow equations has impeded the progress of CFD to a very large extent until the TVD (total-variation-diminishing) algorithms were developed in the late 1980s and early 1990s. As a result, accurate numerical methods were available to resolve the complex wave interactions in multi-dimensional flows. At the same time, giga-flop computing powers also became available for computational fluid dynamics (CFD) simulations. Therefore, many SRM ignition transient CFD simulations were published in the 1990s.[3, 5-8] Prediction methodologies have been also extended to include the coupling of flowfield and propellant deformation.[9-11] However, most of the developments in this field are still based on the work of Peretz, et al.[2] with the heat fluxes being computed by using a variety of semi-empirical formulas. It is fair to say that, at the present, progress is hindered by the lack of reliable physical models for heat transfer. Fortunately, as indicated by Wang et al[9] and demonstrated later in this paper, the computation of heat transfer in a CFD code can be conducted separately in a subroutine which can be replaced or updated without the need to change other parts of the code. Due to the progress in the performance enhancement of space launch systems, star- and fin-shape propellant grain designs are now being used frequently. Therefore, there is an urgent need to replace 1-D and 2-D approaches by 3-D methodologies.

Recently, funded by DOE's Accelerated Strategic Computing Initiative (ASCI) project, the University of Illinois has established a Center for Simulation of Advanced Rockets (CSAR).[12] The goal of the center is to develop "an integrated rocket simulation tool capable of detailed, whole-system simulation of solid-propellant rockets under both normal and abnormal operating conditions." A report on the Center's progress was presented by Alvilli et al. [8]

With today's computing power, enhanced by visualization software, 3-D simulations of the SRM ignition transient are tractable. However, depending on the size of the burning area to be simulated and the computational grid resolution used, it could be very time consuming. The reason is that the mass injection on the burning surface belongs to a subsonic inflow boundary condition in CFD. An iteration procedure is therefore required, which is time consuming due to the non-linear coupling of the equations. For the simulations presented in this paper, a quasi-3D approach is applied. Each case takes 36 CPU-hours on a Cray-SV1. This running time is considerably less than what is expected for a full 3-D simulation.

The computation can be sped up by using parallel computing systems. As an example of speed-up, super parallel computers have been applied by Wang and Taylor [13] to simulate the transonic flowfield over a complete plume-on Delta II 7925, which has one core vehicle plus 9 strap-on SRM boosters. Using 506 nodes of an Intel's Paragon, a converged solution can be obtained in 48 hours. It was estimated that it would have required at least three months for the simulation if it had been done on a vector machine, Cray-YMP. There is no doubt that future 3-D SRM ignition transient simulations will rely on super parallel computers. As reported by Alvi et al,[8] super parallel computers are being used in CSAR at the University of Illinois. However, at present, full 3-D CFD simulations using parallel computers are still not popular. Furthermore, to convert an existing 3-D CFD code to run on a parallel computing system is not an easy task.[14] A quasi-3D approach may be considered as an alternative cost-saving approach.

In this paper, the physical models currently used in CFD for SRM ignition transient are reviewed. A quasi-3D flowfield simulation methodology for SRMs with large numbers of fins will be presented. The first objective is to show the 3-D phenomena in the flowfield that could not be simulated using a 1-D or 2-D CFD approach. The second objective is to illustrate an easy procedure that could lead to parallel CFD implementation. In this quasi-3D approach, an axi-symmetric system of equations is applied to the bore and a 2-D Cartesian system of equations to the fins. The fundamental assumption is that the circumferential, θ-component, velocity in the bore region, and the w-component velocity, which is normal to the fin surface, in the fin region are negligible. The justification is that, as the number of fins increases, the planes of symmetry increase and the momenta associated with these velocity components cancelled each other on the plane of symmetry. In this paper, the flow will be assumed inviscid and the simplest heat transfer and ignition models will be applied. Since the purpose of this paper is to illustrate the methodology and the flow physics that the present approach can bring forth, generic SRMs will be used.

Review of Models

A CFD simulation of SRM ignition transient involves the modeling of flowfield, heat transfer, ignition, and burning surface.

Ignition Model

The propellant surface temperature rises due to heat flux from the hot gas flowfield adjacent to it. A part of the heat flux is conducted into the propellant. According to Peretz et al,[2] a point on the surface ignites when the temperature at that point attains an ignition temperature. To determine the surface temperature, a 1-D unsteady heat conduction equation was applied in a semi-infinite domain with a constant temperature as initial condition and an unsteady heat flux as a boundary condition. The equation is solved simultaneously with flowfield equations in a coupled manner. In an alternative approach, Johnston[6] approximated the solution by uncoupling this equation from the flowfield equations. When the conduction equation is uncoupled from the flowfield equations, an exact solution is given in Churchill:[15]

$$T_s(t,\bar{x}) = T_p + \sqrt{\frac{\alpha}{\pi}}\int_0^t \frac{q(\tau,\bar{x})}{k}\frac{d\tau}{\sqrt{t-\tau}} \qquad (1)$$

where,

$q(t,\bar{x})$ = heat transfer rate to the propellant surface located at $\bar{x}$ and at time t

T_p = propellant initial temperature

T_s = propellant surface temperature

α = propellant thermal diffusivity

k = propellant thermal conductivity.

Using this approach, it is required to store $q(t,\bar{x})$ for all time steps of computation. For multidimensional computations, the approach requires a large amount of computer memory space. To ease this burden, Alavilli et al[8] and Luke et al[3] have presented methods with further simplifications.

Heat Transfer Model

The heat flux $q(t,\bar{x})$ to the propellant surface includes convective heat flux $q_c(t,\bar{x})$ and radiative heat flux $q_r(t,\bar{x})$.

$$q(t,x) = q_C(t,\bar{x}) + q_r(t,\bar{x}) \qquad (2)$$

An exact formulation to compute $q(t,\bar{x})$ is highly desirable. It is, however, a very difficult task. The use of semi-empirical formulas is the current state-of-the-art and all the papers published so far have used this approach. Although there are only a few papers published on this subject, which applied Reynolds-averaged Navier-Stokes (RANS) equations, the evaluation of the heat fluxes to the propellant surface also resorted to empirical equations. The work of Le Helley,[7] was based on the RANS with a k-L turbulent model. However, the convective heat flux was computed based on an ad hoc formula and the radiative heat flux was neglected. The work of Ciucci et al[5] applied a k-ε two-equation turbulent model. Since the paper addressed the transient phenomena induced by igniters only, heat transfer computations were totally neglected.

For the convective heat flux, q_c, both Johnston[6] and Luke et al[3] applied semi-empirical formulas. Johnston[6] applied a convective heat transfer coefficient for turbulent flow inside tubes and ducts given in Section (8-3) of reference (19):

$$q_c(t,\bar{x}) = C_c Pr^{-2/3} c_p (\mu/L)^{0.2} (\rho\,|\,v\,|)^{0.8} \times (T_{ad} - T_s) \qquad (3)$$

where Pr is the Prandtl number; c_p, the specific heat at constant pressure; L, a reference length; C_c, an empirical constant; $|v|$, the magnitude of the velocity vector on the surface (for inviscid flow); T_{ad} , a recovery wall temperature; and T_s , the wall temperature given by Eqn. (1).

Semi-empirical formulas for the radiative heat transfer applied by Johnston[6] and Luke et al [3] can be generalized for 3-D flows by

$$q_r = (1-\eta)C_{r1}\sigma(T^4 - T_s^4) + \eta C_{r2}\sigma(T_{flame}^4 - T_s^4) \qquad (4)$$

where η is a weighting function, $\eta=\Sigma A_{lm}/(\Sigma A_{lm}+A_{ij})$; A_{lm} , the ignited computational cell surface area adjacent to the surface (i , j) with area A_{ij}; σ, the Stefan-Boltzmann constant; C_{r1} and C_{r2} , view factors. C_{r1} and C_{r2} can be computed accurately from the equations given in Section (5-6) in reference (19).

CFD Model

For the inviscid flow, the Euler's equations can be written in a conservation law form as,

Continuity equation:

$$\partial(\rho y^n)/\partial t + \partial(\rho u y^n)/\partial x + \partial(\rho v y^n)/\partial y = 0 \qquad (5)$$

Momentum equations:

$$\partial(\rho u y^n)/\partial t + \partial[(p+\rho u^2)y^n]/\partial x + \partial(\rho u v y^n)/\partial y = 0 \qquad (6)$$

$$\partial(\rho v y^n)/\partial t + \partial(\rho u v y^n)/\partial x + \partial[(p+\rho v^2)y^n]/\partial y = np \qquad (7)$$

Energy equation:

$$\partial(\rho e y^n)/\partial t + \partial[\rho u(e+p/\rho)y^n]/\partial x + \partial[\rho v(e+p/\rho)y^n]/\partial y = 0 \qquad (8)$$

where n = 0 or 1 for a 2-D or an axisymmetric flow, respectively; ρ, the gas density; p, the pressure; x and y, spatial independent variables in the axial- and radial- directions, respectively; u and v, the velocity components in the x- and y- directions, respectively; and e, the total energy per unit mass. Assuming a perfect gas, e is related to p by the equation of state

$$p = (\gamma - 1)\rho[e - (u^2 + v^2)/2]. \qquad (9)$$

In the 1-D or 2-D approach, the chamber of the star-shaped propellant grain was approximated by a cylinder of equivalent volume and surface. In the work of Luke et al[3], 1-D Euler's equations were used. The effect of fin/star-shape was approximated by plume expansion. This work identified the cause of the high rise rate of the Space Shuttle's Redesigned Solid Rocket Motor (RSRM) head end chamber pressure. The paper also illustrated that a volume-filling approach, if applied properly, could be used to predict the chamber pressure rise in a large L/D motor. Johnston[6] applied an axisymmetric flow code developed by Wang et al.[16]

Burning Surface Model

Propellant burning physics is coupled to the flowfield by its burning rate and flame temperature. Neglecting the erosive burning, an idealized propellant burning rate equation can be written as:

$$\dot{r} = ap^{\kappa} \qquad (10)$$

where $\dot{r}$, the burning rate, has the dimension of length/time; a, an empirical constant; κ, a constant, known as the burning rate pressure index. Luke[3] et al applied a more elaborate burning equation.

Eqn. (10) enters the flowfied solution in the form of a subsonic mass flux into the computational domain. The mass flux rate, $\dot{m}$, is given by

$$\dot{m} = \rho_s v_n = \rho_p \dot{r} \tag{11}$$

where ρ is the gas density on the burning surface; v_n, the velocity normal to the surface; and ρ_p, the propellant density. Using a finite volume approach for the flowfield, the pressure and density on the surface need to be evaluated. Both are obtained by solving two simultaneous nonlinear algebraic equations. One of the equations comes from the Riemann invariant, and the other from the invariant of total enthalpy. Details of the equations are given in reference (18).

A Quasi-3D Flowfield Approach

Methodology

The methodology can be illustrated by considering the generic motors shown in Fig. (1). The motors are similar in that both have large fin areas at the head end region. However, the shapes of the fins are different. Each motor has 7 fins that generate14 planes of symmetry. In a cross section perpendicular to the motor axis, the thickness of the fin is small compared with its radial extent. Also, the area of the bore between any two adjacent planes of symmetry is small compared with the total cross-sectional area of the bore. Under these conditions, the lateral velocity components, w-component and θ-component, are expected to be small in comparison with the axial- and radial- velocity components. This is the main justification for the present quasi-3D approach.

The computational grids are generated for the xy-plane, which lies on the middle of the fin slot. Three grid blocks are used. Block (1) includes the bore, the converging-diverging nozzle, the expansion cone, and the external region. Block (2) is generated for the fin only. Block (3) encompasses the external region of block (1). Boundary conforming algebraic grids are used.

The transient flowfield simulations are performed using a multi-block version of the ALSINS (Aerospace Launch Systems Implicit/Explicit Navier-Stokes) code.[14] The code was developed based on a finite volume TVD algorithm.[16] The flow equations are solved in a transformed domain. For blocks (1) and (3), the metric coefficients are computed based on a cylindrical coordinate system and the flow equations are solved in axi-symmetric form (n = 1). For block (2), the metric coefficients are computed based on a Cartesian coordinate system and the flow equations are solved in 2-D flow form (n = 0).

On the interface between blocks (1) and (2), as well as that between blocks (1) and (3), the fluxes are constructed based the conservative variables, density, momentum, and specific total energy, at the two cell centers on each side of the interface. As shown in Fig. (1c), the upper surface of the last cell in the y-direction of block (1) has two parts. One part is connected to block (2) and is treated as an interface. The other part is treated as a solid boundary before it is ignited and as a burning surface after it is ignited.

Discussion of Results

The transient flowfield is driven by a specified input for igniter mass flow rate and enthalpy. Some highlights of the transient characteristics that may not be attainable by using an axi-symmetric approach will be discussed here. Fig. (2a) shows the head end chamber pressure. The fin areas of the two configurations are close but not equal. Consequently, the steady state pressures are not equal. The overall pressure rise rates are similar. Fig. (2b) and Fig. (2c) show the head end pressure together with the centerline pressure at a few locations as marked in Fig. (1). The igniter shock, marked by "A" in figures (2b) and (2c), as well as its propagation can be clearly identified. The trace of the compression wave reflected from the converging part of the Laval nozzle is also shown, marked by "B" in figures (2b) and (2c). It is seen that configuration 2 renders a higher level of pressure oscillations inside the SRM than configuration 1 does. Shown in Fig. (3), Fig. (4), and Fig. (5) are the comparisons of temperature, burning surfaces, and velocity vectors, respectively, at t = 60 ms. The effects of the fin configuration are clearly shown in the figures. Large reverse flow regions are shown in Fig. (5). The detailed velocity variations have important impacts to the system designs[4] and can not be obtained using a 1-D or a 2-D simulation.

Parallel Computing

A procedure for the development of a parallel CFD code was given in Reference (14). The procedure applied a domain decomposition by which a computational domain is decomposed into blocks and each block is assigned to a processor. On the interface, the neighboring blocks exchanged information necessary for the construction of numerical fluxes as mentioned earlier. The procedure becomes transparent to the code users if the block interface is treated as one of the boundaries. This procedure is applied to the simulations presented in this paper although the code was run on a CRAY-SV1, which is a vector machine. A simple instruction using MPI can exchange the information, as indicated in Reference (14). The exchange of information can be done in a subroutine.

The difficulty of long running time due to the iteration procedure is necessitated by the modeling of the burning surface and can be surmounted by load balancing procedures. The simplest procedure, static load balancing, is to assign the processors fewer cells with burning surfaces, and more cells that have no burning surfaces.

Summary and Conclusion

A review of the current state-of-the-art in solid rocket motor (SRM) ignition transient prediction has been presented. The lack of reliable physical models is a hindering factor to progress in this field. The currently available CFD methodology and computing power have made the development of 3-D SRM ignition transient codes tractable, but at a high cost in terms of long running times and complexity. A quasi-3D CFD approach has been presented as an alternative cost-saving method for 3-D ignition transient flow predictions. Justifications and limitations have been discussed. The approach was applied to two generic SRMs to illustrate (1) 3-D flow phenomena that can not be obtained by using a 1-D or 2-D approach and (2) an easy route that could lead to parallel CFD.

Acknowledgment

This work is partially supported by Product Development Funds of The Aerospace Corporation. Simulations are performed using DoD Shared Major Computing Resource Center in the US Army Research Laboratories, Aberdeen, Maryland. Figures presented in this paper are prepared by Mr. E Park and W. Austin.

References

(1) Salita, M., "Rocket Propellant Ignition: A Workshop Report," Proceedings of the 22nd JANNAF Combustion Meeting, Vol. I 1985, pp65-86.

(2) Peretz, A., Kuo, K. K., Caveny, L. H., and Summerfield, M. "Starting Transient of Solid-Propellant Rocket Motors with High Internal Gas Velocities," *AIAA Journal*, Vol. 11, No. 12 1973, pp 1719-1727

(3) Luke, G. D., Eagar, M. A., and Dwyer, H. A. "Ignition Transient Model for Large Aspect Ratio Solid Rocket Motor," AIAA paper 96-3273, 32nd AIAA/ASME/SAE/ ASEE Joint Propulsion Conference, July 1-3, 1996 Lake Buena, FL.

(4) Kumar, M., and Kuo, K. K., "Flame Spreading and Overall Ignition Transient," Progress in Aeronautics and Aeronautics, AIAA, Vol. 90, K. K. Kuo Edt, 1984.

(5) Ciucci, A., Jenkins, R. M., Foster, W. A. "Numerical Analysis of Ignition Transients in Solid Rocket Motors," AUAA/SAE/ASME 27th Joint Propulsion Conference, Jun 24-26 1991, Sacramento, CA.

(6) Johnston, W. A. "Solid Rocket Motor Internal Flow During Ignition," AIAA paper 91-1655, also *AIAA Journal of Propulsion and Power*, Vol. 11, No. 3, 1995, pp489-496.

(7) Le Helley, P. "3D Turbulent Navier-Stokes Simulations of Ignition Transients in Solid Rocket Motors," AIAA Paper 98-3843, 34th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, Cleveland, OH, July 13-15, 1998.

(8) Alvilli, J. Buckmaster, J., Jackson, T. L., and Short, M., "Ignition-transient Modeling for Solid Propellant Rocket

Motor," AIAA paper 2000-3567, 36th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exibit, July 16-18, 2000, Huntsville, AL

(9) Wang, J. C. T., Yang, S. H., Than, P. T., and Ndefo, E. D.,"Coupled Transient Flowfield and Propellant Deformation Analysis for the Titan IV 7-Segment Solid Rocket Motor," AIAA Paper 94-3285, 30th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, June 27-29, 1994, Indianapolis, Indiana.

(10) Chang, I. S., Patel, N. R., and Yang, S. H., "Titan IV SRMU Anomaly and Redesign," 30th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, June 27-29, 1994, Indianapolis, Indiana.

(11) Johnston, W. A., and Murdock, J. W., "Flow-Structural Interaction Inside a Solid Rocket Motor During Ignition Transient," AIAA Journal of Propulsion and Power, Vol. 11, No. 5, 1995,

(12) Heath, M, "Whole-system Simulation of Solid Rockets Is Goal of ASCI Center at Illinois," SIAM News Volume31, No. 4, May 1998.

(13) Wang, J. C. T., and Taylor, S., "Large Scale Navier-Stokes Multi-Body Launch Vehicle Flowfield Simulations," Proceedings, 20th Congress of International Council of the Aeronautics, Sorrento, Italy, 8-13 Settember, 1996.

(14) Wang, J. C. T., and Taylor, S., "An Architecture-Independent Navier-Stokes Code," Chapter 10, Industrial Strength Parallel Computing, A. E. Koniges edt., Morgan Kaufmann Publishers, San Francisco, 2000, pp147-167.

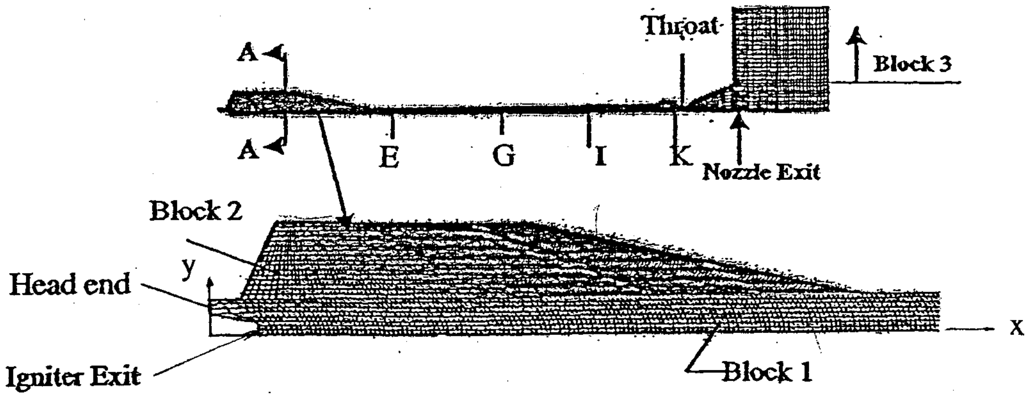
(15) Churchill, R. V., "Operational Mathematics," 3rd Ed., Section 47, McGraw-Hill, 1958, p150.

(16) Wang, J. C. T. and Widhopf, G. F., "A High-Resolution TVD Finite Volume Scheme for the Euler Equations in Conservation Law Form," AIAA Paper 87-0538; also, Journal of Computational Physics, Vol. 84, N0. 1 1989, pp145-173.

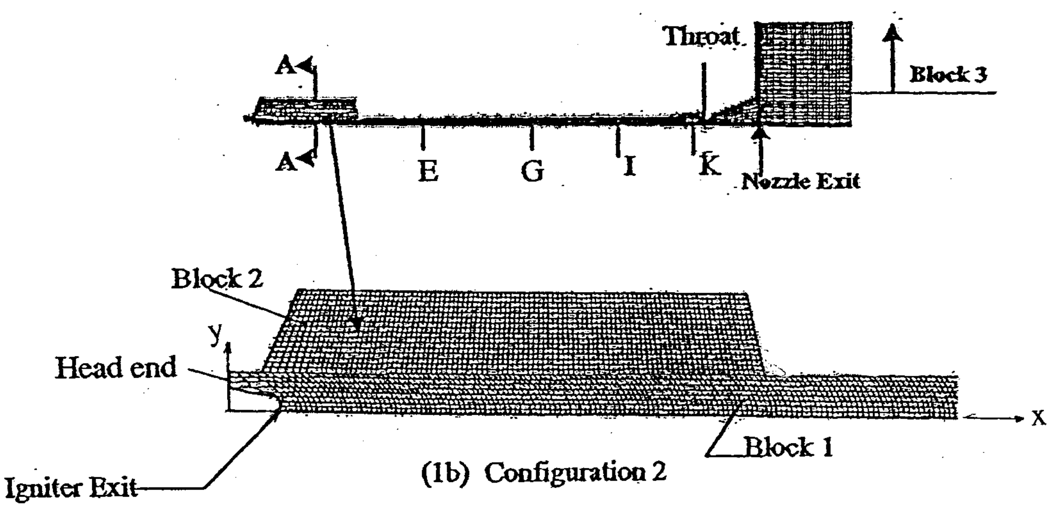
(17) Sutton, G. P., Rocket Propulsion Elements, John Wiley &Sons, Inc. New Yprk, 1992, p375.

(18) Ton, V. T. Wang, J. C. T., and Widhopf, G. F., "Segmented Solid Rocket Motor Internal Flow Simulations," AIAA Paper 90-0683, 1990.

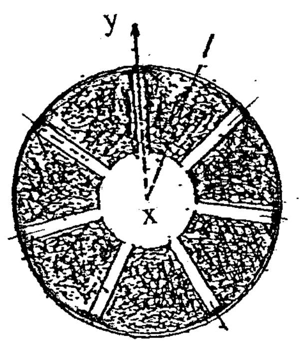
(19) Keith, F., Principles of Heat Transfer, 2nd Edition, International Textbook Co., 1965.



(1a) Configuration 1



(1b) Configuration 2



(1c) Cross-section A-A

Fig.(1) Generic SRM configurations and computational grid systems

Fig. (2a) Comparison of Head End Chamber Pressure

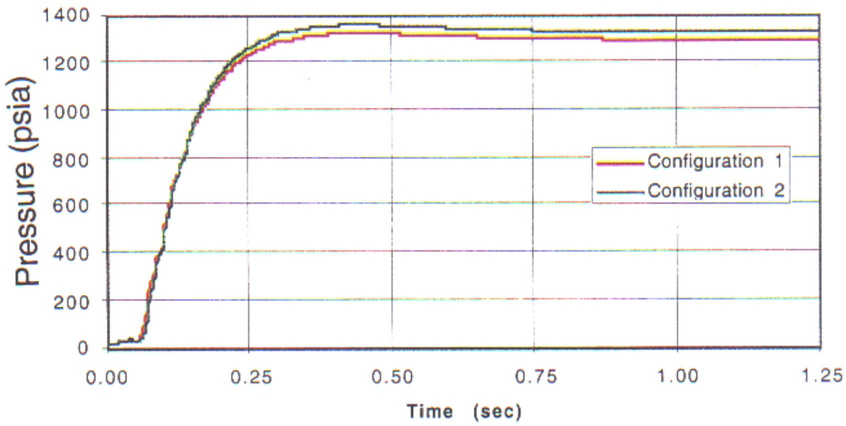


Fig. (2b) Ignition Shocks and Compression Waves in Configuration 1

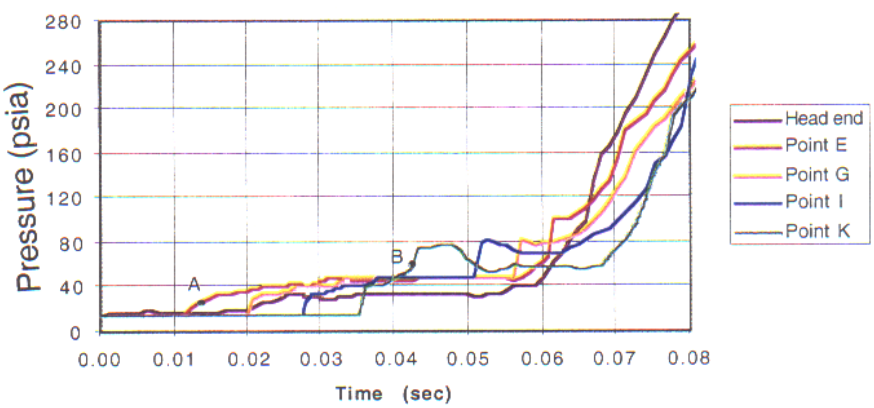
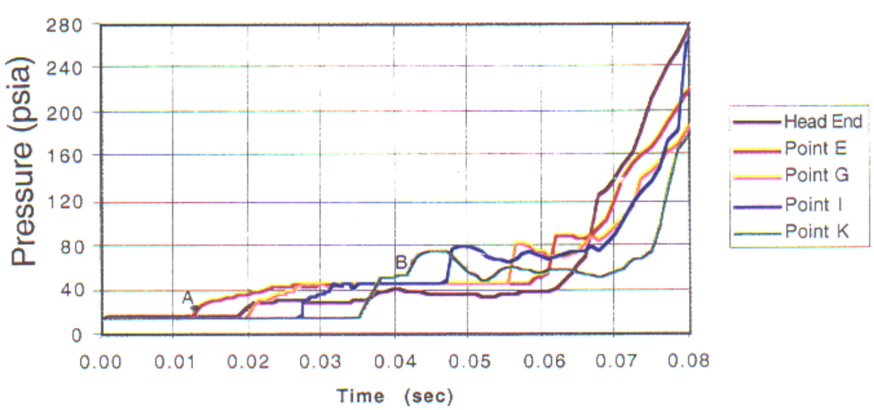


Fig. (2c) Ignition Shocks and Compression Waves in Configuration 2



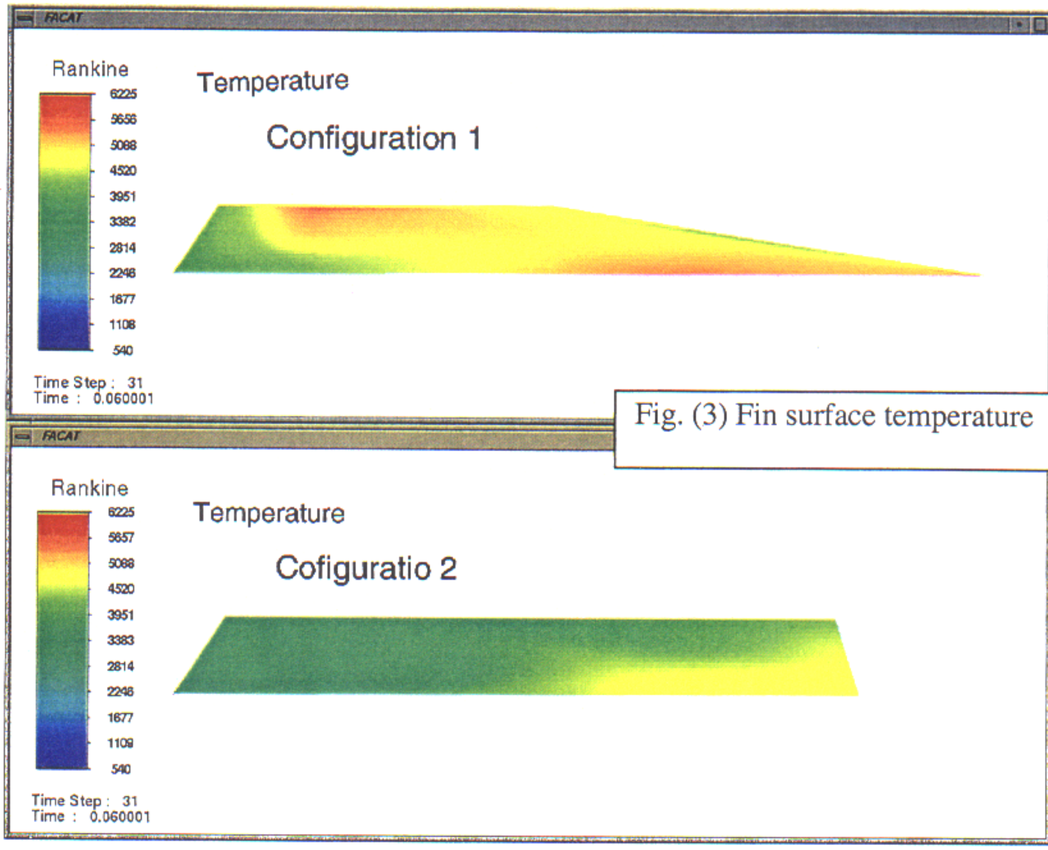


Fig. (3) Fin surface temperature

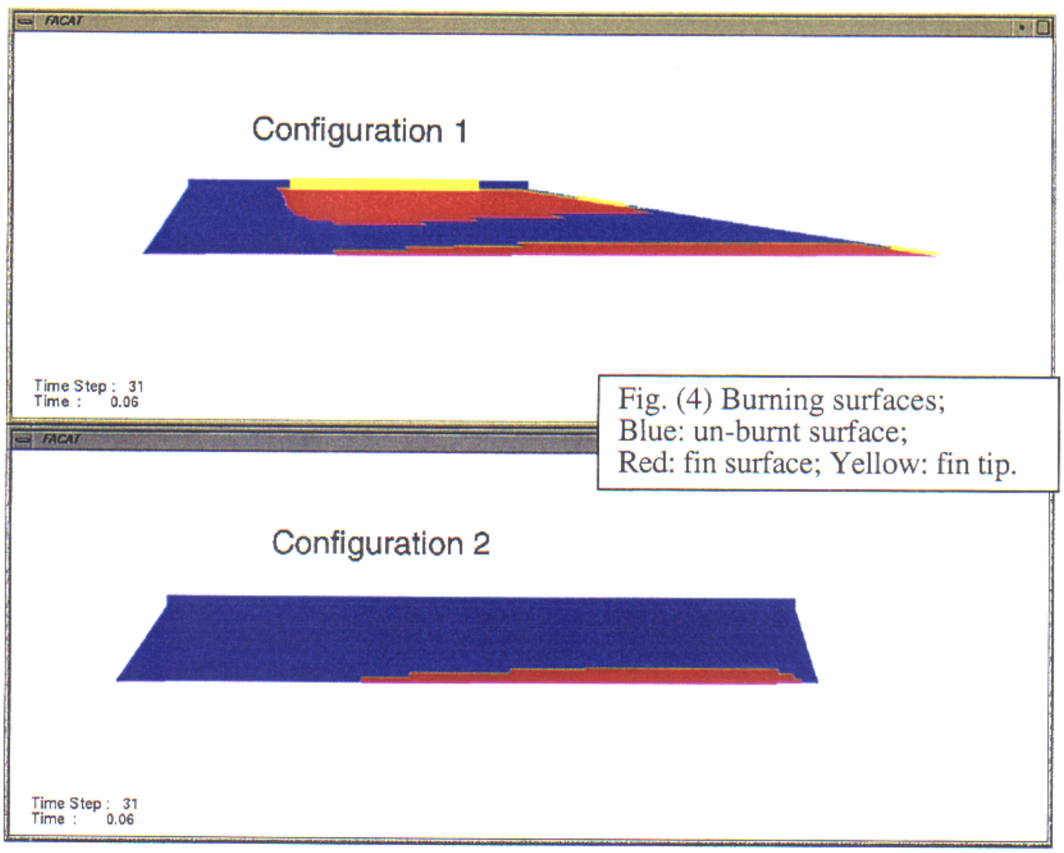


Fig. (4) Burning surfaces;
Blue: un-burnt surface;
Red: fin surface; Yellow: fin tip.

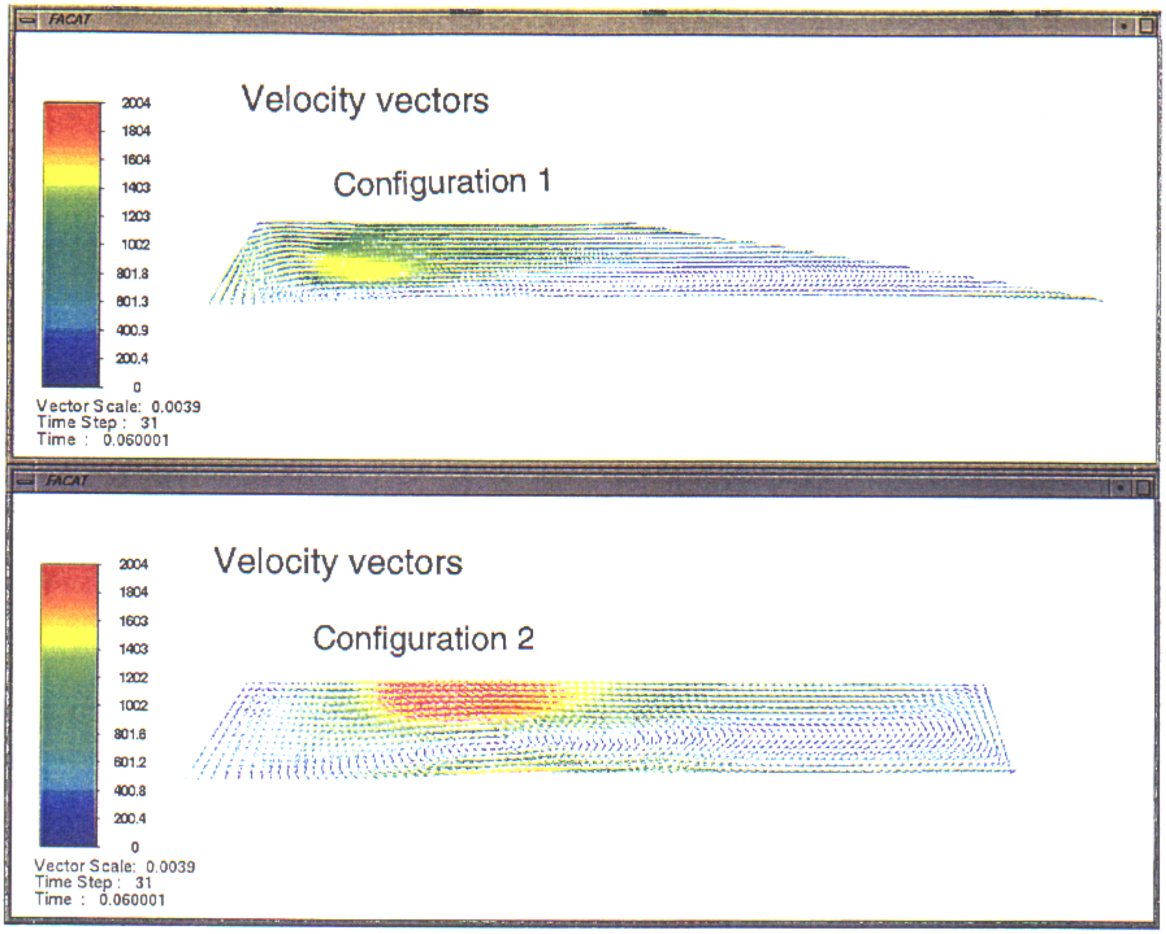


Fig. (5) Velocity vectors: The color shows relative speed